

Overview

PIs Napolitano (structural vulnerability and community co-design), Wauthier (SAR processing and uncertainty quantification), and Busse (hydrological modeling and mass balance), all at Penn State, propose a closed-loop cyber-physical system (CPS) for flood duration estimation, pre-code building vulnerability assessment, and community retrofit decision support in small riverine communities. The small, aging communities most repeatedly devastated by riverine flooding lack the data, software, and technical capacity that existing flood damage frameworks require. Community partners in Montpelier, VT have identified inundation *duration*—not just peak flood stage—as the decision-relevant variable motivating every modeling constraint in this proposal.

The proposed system has four coupled layers. The **physical layer** is the flood-building interaction: pre-code load-bearing masonry and timber structures absent from every existing vulnerability framework. The **sensing layer** is multi-temporal SAR imagery—open Sentinel-1 C-band for flood-extent context and commercial ICEYE X-band for building-level duration and depth at sub-daily cadence. The **cyber layer** is a Bayesian fusion framework integrating SAR-derived observations with hydrological mass balance into probabilistic duration estimates with propagated uncertainty. The **decision layer** is a multi-criteria retrofit allocation framework, co-developed with community stakeholders, that maps duration outputs to building-specific damage probabilities, placing community-scale drainage investments and individual building retrofits on the same probabilistic basis. The loop closes at three timescales—pre-storm staging from NOAA forecasts, post-event updating of priors and error models, and long-term retrofit feedback through drainage and fragility updates—with human decisions as the actuators, making the community a functional component of the control loop.

Intellectual Merit

The intellectual merit rests on two contributions. First, we establish that community-accessible sensing, fused under a Bayesian architecture, can substitute for resource-intensive physics-based simulation in closed-loop CPS decision systems—a generalizable principle for infrastructure contexts without hydrodynamic modeling capacity. Second, we advance a co-design methodology that treats locally held knowledge as structural input to model calibration and characterizes whether community knowledge functions as a substitute for or complement to national geospatial data, providing the first empirical basis for deciding when elicitation improves CPS accuracy. Two Montpelier flood events (2023, 2024) provide a within-community train-test validation: the inference chain is calibrated on 2023 and validated against the held-out 2024 event, directly testing generalizability across event characteristics. ICEYE commercial SAR provides building-level flood depth and duration at sub-daily cadence for 271 downtown buildings across both events, establishing the empirical foundation for fragility calibration and held-out validation.

Broader Impacts

The framework enables duration-based damage assessment where no path currently exists, resolving the allocation problem—community-scale drainage versus individual building retrofits—that annually re-injures small historic municipalities. The decision layer and co-design process are designed for communities without ongoing research support. One postdoc and one PhD student receive cross-disciplinary training in SAR remote sensing, probabilistic structural modeling, hydrological mass balance, and participatory community research. The Community-Inspired Research (CIR) co-design methodology, documented as a transferable protocol, establishes how resident knowledge anchors model calibration and community decisions feed back into physical infrastructure sensing. The framework will be released under an MIT license, enabling communities beyond Montpelier to apply duration-based assessment using freely available national data.